Classical Particle Mechanics as a Suppes Predicate

Formalization notes

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Abstract

This note expresses the axiomatic system for classical particle mechanics of McKinsey, Sugar, and Suppes as a set-theoretic predicate, and gives a Lean 4 encoding of its model structure. It distinguishes the six axioms from the additional results discussed by the original authors. The Lean file formalizes the axiom schema over an explicit abstract signature; it does not implement real analysis or prove the original paper's embedding theorems.

1 The Suppes-style predicate

Fix an ambient signature consisting of a real time domain T, a finite-dimensional real vector space V, a force-index set I, and the usual notions of interval, positive real, twice differentiable trajectory, absolute convergence, and vector sum. A candidate system is the tuple

$$\mathcal{M} = (P, T, m, s, f),$$

where P is the particle set, $m : P \to \mathbb{R}$ is mass, $s : P \times T \to V$ is position, and $f : P \times T \times I \to V$ gives the individual forces. Its predicate is

$$\mathrm{CM}(\mathcal{M}) \quad \Longleftrightarrow \quad P_1 \wedge P_2 \wedge P_3 \wedge P_4 \wedge P_5 \wedge P_6,$$

with the clauses in Table 1. This is a definition of which structures count as models, not a claim that every arbitrary tuple satisfies the laws.

2 Reading the axioms and the reported results

The kinematic and dynamical clauses work together: P_3 supplies acceleration as the second time derivative of position, P_5 makes the total force

	Axiom	Condition
P_1	Particles	P is finite and nonempty.
P_2	Time	T is an interval in $\mathbb{R}$.
P_3	Kinematics	For each $p \in P$, $t \mapsto s(p,t)$ is twice differentiable on T.
P_4	Mass	For each $p \in P$, $m(p) > 0$.
P_5	Forces	For each $(p,t) \in P \times T$, the series $\sum_{i \in I} f(p,t,i)$ is absolutely convergent.
P_6	Dynamics	For each $(p,t) \in P \times T$, $$\sum_{i \in I} f(p,t,i) = m(p)\,\frac{\partial^2 s}{\partial t^2}(p,t).$$

Table 1: The axioms defining an n-dimensional system of particle mechanics. The vector space V is $\mathbb{R}^n$.

well-defined independently of the order of summation, and P_6 equates that force with mass times acceleration. The original paper does not assume Newton's first or third law as axioms. It reports the first law as a consequence of its system and proves an embedding result relating its models to Newtonian systems; those results are not formalized in the Lean file here [2].

In the Suppes approach, a set-theoretic predicate is a formula on structures that selects the models of a theory. Da Costa and Krause emphasize that transportability is relevant when the intent is to capture all structures of a kind, but need not be imposed when a predicate is intended to select only a particular class [1]. Accordingly, CM is stated relative to the ambient signature and to the chosen class of particle systems; no claim of transportability is made here. Suppes's broader account of scientific structures is a methodological reference for this representation [3].

3 Lean encoding and scope

The companion file `_lean/ClassicalMechanics.lean` defines a signature carrying the semantic operations and predicates for time intervals, positive masses, differentiability, summability, and force summation. A `System` supplies particles and the functions m, s, acceleration, and f, together with

witnesses of P_1–P_6. The theorem `finiteNonemptyParticles` packages P_1, and `secondLawHolds` exposes P_6 as a checked Lean theorem.

This is a typed, machine-checked formalization of the structural axiom schema, not a development of calculus over $\mathbb{R}$. The interpretation of the signature's predicates and operations must be supplied by a mathematical library or a concrete model before numerical or analytic consequences can be proved. The result is therefore deliberately narrower than a full formalization of the cited paper.

References

[1] Newton C. A. da Costa and Décio Krause. Suppes predicate for classes of structures and the notion of transportability. Preprint, PhilSci-Archive, eprint 17445, 2020.

[2] J. C. C. McKinsey, A. C. Sugar, and Patrick Suppes. Axiomatic foundations of classical particle mechanics. *Journal of Rational Mechanics and Analysis*, 2(2):253–272, 1953.

[3] Patrick Suppes. *Representation and Invariance of Scientific Structures*. Number 130 in CSLI Lecture Notes. CSLI Publications, Stanford, CA, 2002.